THE UNITED REPUBLIC OF TANZANIA

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KIBAHA PRIVATE SECONDARY SCHOOL ASSOCIATION (KIPSSA)

FORM TWO PRE-MOCK EXAMINATION

032 CHEMISTRY

Time: 2½ Hours **Wednesday, 28th May 2025, p.m.**

Instructions

1. This paper consists of sections **A**, **B** and **C** with a total of ten (10) questions.
2. Answer **all** questions
3. Cellular phones and any unauthorized materials are not allowed in the examination room.
4. Write your examination number on every page of your answer booklet (s)
5. The following atomic masses may be used:

   H=1, N=14, O=16, Na=23, S=32

QUESTION NUMBER	MARKS	INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

SECTION A (15 MARKS)

1. For each of the items (i)-(x) choose the correct answer among the given alternatives and write its letter beside the item number in your answer sheet provided.

(i) Seypalm's child was stung by a wasp. Which of the following would you advise him to apply on the affected area of his child's body?

A. Vinegar
B. Washing soda
C. Ethanol
D. Water

(ii) Which of the following sets of substances contain mixtures only?

A. Water, cooking oil and petrol
B. Steel, bronze and brass
C. Iron, brass and Ice
D. Table salt, sugar and chalk

(iii) Substances that can burn or corrode your skin are called…

A. Oxidizing agent
B. Flammable
C. Corrosive
D. Biohazard

(iv) Neema looked at the periodic table then selected two elements. Element Q of group II and P of group VII. If these elements are chemically combined, the formula of the compound formed is:

A. Q_2P_7
B. QP_7
C. Q_2P
D. QP_2

(v) During welding process, welders prefer to use non-luminous flame for their work simply because:

A. It is readily available
B. Produce very hot flame
C. It can be made by kerosene
D. It is easy to transport.

(vi) Form two students prepared gas Q in the laboratory. They collected gas Q by downward delivery method. This is because:

A. It burns in the air with a pop sound.
B. It can fill balloons
C. It is lighter than water
D. It is more soluble in water

(vii) Esther put an ink on Kikoti's shirt. Kikoti went to the laboratory technician to remove that ink. Which of among the following methods did the laboratory technician use?

A. Filtration.
B. Sublimation
C. Layer separation
D. Chromatography.

(viii) Mrs. Seypalm decided to change from using kerosene as a source of energy to solar energy. When she started using it, the flame was not having the problem like that of the kerosene, this implies that Mrs. Seypalm used;

A. Non-renewable source of energy.
B. Friendly energy sources.
C. Renewable sources of energy
D. Destructive energy sources.

(ix) The following are good examples of chemical change, except:

A. Food scrap turn into compost
B. Melting of candle wax
C. Burning of candle wax
D. Rusting of iron.

(x) During an experiment, dry hydrogen gas was passed through a test tube containing a strongly heated black copper oxide without any air. What was eventually observed in the test tube?

A. The black copper oxide changed into white copper metal
B. The black copper oxide changed into brown copper metal.
C. The hydrogen gas exploded with a pop sound
D. The copper dissolves to form salt

Answers

i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.

2. Match the chemical composition of the portable fire extinguisher in List A with the correct portable fire extinguisher in List B and write its letter in the table provided below.

LIST A	LIST B
(i) Monoammonium phosphate with nitrogen carrier (ii) Non-flammable carbundioxide gas under extreme pressure (iii) Fine sodium bicarbonate powder pressurized by nitrogen (iv) Ordinary tap water pressurized by air (v) Potassium acetate	A. Foam fire extinguisher B. Wet chemical fire extinguisher C. Air pressurized water (APW) fire extinguisher D. Dry chemical fire extinguisher E. ABC fire extinguisher F. Halon fire extinguisher G. Carbon dioxide fire extinguisher

Answers

i.	ii.	iii.	iv.	v.

SECTION B (70 MARKS)

Answer all questions in the space provided.

3. Form two students prepared gas X with the following properties: it is colourless, odourless, tasteless, slightly soluble in water and it's a strong reducing agent. As a form two student:

   (a) Give the name of gas X …….

   (b) Name a compound and a metal that can be used to prepare gas X

   (i) Compound ……

   (ii) Metal ……

   (c) Mention two (2) uses of gas X

   (i) ……

   (ii) ……

   (d) Draw a well labelled diagram showing the preparation of gas X

4. (a) A Chemistry teacher was doing experiments in the laboratory with form two students. He decided to place a large crystal of potassium permanganate in the bottom of a beaker of cold water and left it for several hours.

   (i) Describe what he must observe after five minutes. Explain your answer

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(ii) Describe what he is likely to observe after several hours. Explain your answer

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(iii) Name the two processes which have taken place during the experiment

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(b) Scientists use a short form to express the name of an atom that can be used in chemical reaction, for example, they use a symbol Cl for a certain element. Why do they use this form? Give four reasons.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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5. You have been given elements M and N with atomic numbers of 20 and 9 respectively.

(i) Show how the chemical formula is formed by the combination of element M and N, then write the formula of the compound formed.

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(ii) Mention the type of bond formed by the combination of element M and N.

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(iii) State three characteristics of the compound formed by the type of bond named in 5 (ii) above.

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(iv) Using dots and crosses, show how the named type of bond is formed between M and N.

6. (a) You have been appointed in a village meeting as an expert of Chemistry to explain to the villagers four applications of Chemistry in our daily life on the following fields:

(i) Construction material industry

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(ii) Agriculture

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(iii) Medicine and pharmacy

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(iv) Transportation

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(b) A form one student was asked to mention four professional careers he/she may attain through studying Chemistry. What correct answers can we get from him?

(i) ……

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(ii) ……

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(iii)……

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(iv)……

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7. (a) In 1803, Dalton developed the theory about the atom. However, scientists have modified the ideas put forward by John Dalton concerning the structure of an atom. List any four (4) ideas suggested by these scientists so as to amend Dalton's atomic theory.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(b) Fill in the following table concerning the properties of subatomic particles of an atom.

Subatomic particles	Symbol	Location	Charge	Real mass (g)
Proton	P		+1	
	n	In the nucleus	0	
Electron		Outside the nucleus		1/1840

8. (a) You were given the responsibility of recommending a good fuel for use in your village. What qualities of the fuel would you consider?

(i) ……

(ii) ……

(iii)……

(iv)…….

(b) An experiment was done to measure the heat value of biodiesel. In this experiment, 12 litre of water was heated using 56g of biodiesel. The temperature of water raised from 24.7℃ to 68.5 ℃. Determine heat value of the biodiesel. (Specific heat capacity of water = 4.18 KJ Kg^{-1} K^{-1}, density of water = 1000 kg/m3

9. (a) Mixtures can be categorized as homogenous or heterogeneous, with at least one example, explain each of the two categories of mixtures.

(i) A homogeneous mixture:

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(ii) A heterogeneous mixture:

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(b) When it comes to the separation of mixtures, an appropriate method may need to be used in order to make the mixture separation process successful. One of the methods is chromatography; explain the following about it:

(i) Its definition:

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(ii) Two of its practical application in life:

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SECTION C (15 MARKS)

10. (a) A certain compound contains 25.6% copper, 12.8% sulphur, 25.6% oxygen and the rest being water of crystallization. Calculate the molecular formula of a compound if its molar mass is 250. Use relative atomic mass of copper = 64. Show your working clearly.

(b) For each of the radicals given in the following table, write its formula.

	Radical	Formula
(i)	Nitrate	
(ii)	Carbonate	
(iii)	Phosphate	
(iv)	Dichromate	
(v)	Sulphate	

(c) Calculate the oxidation state of the underlined elements in the following radicals:

(i) $\underline{NH_4^+}$

(ii) $\underline{SO_3^{2-}}$

(iii) $\underline{ClO^{4-}}$

(iv) $\underline{MnO^{4-}}$

(v) $\underline{HSO^{4-}}$